

Kaivalya Mase

B.Tech Mechanical Engineering

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Education

Degree	Institute	Year	CGPA/%
B.Tech Mechanical Engineering	IIT Hyderabad	2027	9.4
B.Tech CSE (Double Major)	IIT Hyderabad	2027	6.5
Class XII	Late PB Jog Junior College	2023	89.83%
Class X	Dr Kalmadi Shamarao High School	2021	98.2%

Experience

Applied Materials India – Atomic Layer Deposition May 2026 – Jul 2026

- Investigated process failure due to flaking of coating from ALD chamber components;
- Reviewed possible simulation approaches for modelling the failure phenomenon;
- Concluded simulation approaches infeasible due to high compute costs, and experimental methods more appropriate;
- Proposed an experimental methodology for quantifying the effect of substrate condition on coating adhesion, targeting a **2x increase** in the preventive maintenance interval, with a post-processing workflow using linear regression.

Projects

STOL Aircraft Wing Design – LAT Aerospace, Inter-IIT Tech Meet 2025 Oct – Dec 2025

- Designed and simulated a proposed blown-wing design for a high-lift STOL application;
- Developed Ansys Fluent simulation workflow for an upper surface blown single-flap wing configuration, modelling the propeller as a momentum source to energise the boundary layer and delay flow separation;
- Performed mesh convergence analysis to establish mesh independence, selecting appropriate mesh size;
- Evaluated four possible propeller-aerofoil-flap configurations, selecting the configuration providing a lift coefficient of $C_L = 6.5$ at takeoff, and a lift-drag ratio of $L/D = 5$ at takeoff and $L/D = 20$ at cruise;
- Presented the proposed solution at the Inter IIT Tech Meet, finishing **4th** among 23 participating IITs.

Formula Student – Chassis Team Lead May 2023 – Apr 2026

- Designed and simulated a Formula Student space-frame chassis frame to limit the front-impact deformation to approximately **4 mm** and achieve a torsional stiffness of **1150 Nm/deg** under cornering loads;
- Manufactured the chassis frame using AISI 1010 steel tubes via tube notching and arc welding;
- Coordinated with Vehicle Dynamics and Powertrain teams for integration of suspension and drivetrain assemblies.

Stress Intensity Factors for Brittle Fracture Apr 2026

- Automated Abaqus simulations for **800** instances of the disc-shaped compact tension specimen using Python;
- Generated meshes using **Barsoum quarter-point elements** to model stress singularity at the crack tip;
- Extracted stress intensity factors using the **contour integrals**, validating against the analytical solution;
- Utilised the obtained data to train a linear regression model for predicting the shape function.

1-Dimensional Finite Element Method Solver Oct – Nov 2026

- Formulated the weak form for the Euler-Bernoulli beam bending differential equation with loads and boundary conditions fed through an input file along with the element stiffness matrices and force vectors;
- Implemented a Finite Element Method based code to compute and plot the deflection, shear force, and bending moment;
- Evaluated the convergence in results as a function of the number of elements and order of the shape functions.

Technical Skills

CAD & Modelling	SolidEdge, Fusion360, SolidWorks, Siemens NX
Simulation	ANSYS, Abaqus, MSC ADAMS
Programming	C, Python, MATLAB
Other	Bambu Studio, Creality Print, LaTeX

Relevant Coursework

Mechanics	Finite Element Methods, Fracture Mechanics, Composite Structures and Materials
Manufacturing	Additive Manufacturing, Industry 4.0